

Alterface!

Either you're working with a keyboard and mouse, or reading what your computer displays on-screen. Don't you just wish user interfaces could go through a full makeover?



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We've always been using boring-keyboard and boring-mouse to interact with computers. Why? Because that's what was in use when the PC gained in popularity. As if the case with PCs wasn't bad enough, some mobile devices have incorporated the QWERTY keyboard. You get tiny little keys that are generally used by the thumbs, which the keyboard layout wasn't designed for in the first place. Should we settle for mediocre (or worse) just because it's popular? When it comes to mass-produced devices, sadly, the answer is We Must.

What Choice Do We Have?

For now, none! But more and more innovative research is being done in the field of Human Computer Interaction (HCI), in fields as diverse as touchscreens, natural language processing, eye tracking, and more.

Before you think of questioning the need for new interfaces, think of the popularity of prod-

ucts such as every tablet PC, every PDA, the Wii, and the iPhone. These are popular only because they give you an alternative to pressing buttons or tapping away at a keyboard, and it's mainly their interfaces that set them apart from the crowd.

The GUI

The WIMP interface, short for Windows, Icons, Menus and Pointer, was revolutionary for its time. In 1984 the Macintosh was the first publicly-available computer; it featured the WIMP interface. Twenty-three years later, things haven't changed much, have they?

Now we're definitely not suggesting that interfaces change for the sake of change. The fact is as long as we're limited to a two-dimensional screen as the PC's output, we're probably not going to see anything but the WIMP interface. But with so much research happening to bring 3D realism to our screens, what happens when it does get here? Do you really want to be sitting and trying to use a mouse to drag a 3D window?

Another aspect to consider is the types of tasks we're trying to accomplish: for example, why bother taking word processing to a 3D level? It's not like you will be able to type along all three axes! It's quite obviously the application that needs to be looked at, and one common GUI for everything might not be the way ahead.

Take gaming for example. There are games that are built for consoles and those built for PCs. A racing game, for example, might be a lot easier to play on a console, while an FPS might only be playable on the PC. However, platforms be damned, because both the racing game and the FPS could be so much better if they were more immersive and in 3D. Also, the racing game is a lot more fun if you have a steering wheel control, while an FPS might be more fun if we had a controller that looked like a gun and we had to physically turn around to see what's behind us.

Touchscreens

From PDAs to Tablet PCs and now smartphones, everything is going the touch input way. If we're going to have to use the point-and-click interfaces, nothing beats just tapping a screen—especially